2022 UT Invitational

10/29/2022

Dynamic Planet C Exam Booklet

Parameters

- Each team may bring one binder of any size, and two stand-alone calculators of any type.

Directions

- You will have **50 minutes** to complete this exam - do not begin until instructed to do so!
- All answers must be indicated on your answer document. **DO NOT WRITE ON THIS EXAM!**
- Point values are indicated for all questions & tiebreakers are indicated in test booklets. The number of tiebreakers correctly answered indicates your tie order.
- The maximum number of obtainable points on this exam is 100 points.
- You may take apart the test to split the work, but make sure to staple it back together at the end of the event.

Authors
Lalith Roopesh, ***University of South Florida***
Crystal Nguyen, ***University of Texas - Austin***

(any questions after the conclusion of this event may be emailed to lalithroop@gmail.com and crystal@atxscioly.org)

*** CLASS SET: DO NOT WRITE ON THIS EXAM ***

Refer to Figure A in the images sheet to answer the following questions.

1. This map shows a variety of freshwater features defined by topography. Which of the following is most responsible for the specific erosional patterns shown? (1)
    a. Eolian erosion
    b. Human mining
    c. Glaciation
    d. None of the above have occurred significantly in the area
2. Observe Lincoln Lake. Which of the following terms would best describe Lincoln Lake? (1)
    a. Tarn lake
    b. Paternoster lake
    c. Kettle lake
    d. Supraglacial lake
3. The stream southwest of Lincoln Lake, named Lincoln Creek, is heading in what direction? (1)
    a. Northeast
    b. Southwest
    c. Straight upwards (into the sky)
    d. Impossible to tell
4. Assuming Lincoln Creek runs 10,000 ft in distance (not displacement), what is the approximate gradient of Lincoln Creek shown? (2)
    a. 2%
    b. 4%
    c. 8%
    d. 16%

5. Assume Lincoln Creek runs 9,000 ft in displacement only (straight-line distance) and averages 15 ft of width and 3 ft of depth. Calculate its sinuosity. Show your work. (2)

6. Assuming the stream is in fluvial equilibrium, how would the characteristics of the stream and bed change in the stream southwest of Lincoln Lake as distance from the lake increases? Select all that apply. (2)
    a. The streambed would generally have larger rocks
    b. The stream would generally have larger velocity
    c. The stream would generally carry more sediment
    d. The stream would generally carry finer sediment
    e. The streambed is more likely to be bedrock than sedimentary
7. What is the Strahler order number of Lincoln Creek? (1)
    a. 1
    b. 2
    c. 3
    d. 4
8. What drainage pattern is best shown by the area around Lincoln Lake? (1)
    a. Radial
    b. Dendritic
    c. Parallel
    d. Trellis
9. Using the map and data from previous questions, what type of stream is this? (1)
    a. Braided
    b. Meandering
    c. Straight
    d. Not enough information to determine

10. Near Lincoln Lake, sharp angular rocks are commonly found, in contrast to the small smooth rocks more characteristic of lakes and streams. Explain where these rocks are from and how they likely eroded to that shape. (3)

11. If channel efficiency were to reduce due to bed conditions, how would you expect the following characteristics to change, if at all? (3)
    a. Hydraulic radius
    b. Wetted perimeter
    c. Channel roughness
    d. Average velocity
    e. Capacity
    f. Competence

12. For convenience, a cross-section of Lincoln Creek can be modeled as a rectangle 15 ft in width and 3 ft in depth. Using this model:
    a. Calculate the wetted perimeter of Lincoln Creek (1)
    b. Calculate the hydraulic radius of Lincoln Creek (2)
    c. Experimental markers indicate that the average velocity of Lincoln Creek is 30 ft/s. Calculate the Chézy coefficient for Lincoln Creek in $ft^{1/2}/s$. Show work! (3)
    d. At which part of the stream (upstream closer to Lincoln lake or downstream farther away) would this model apply better? Explain why. (3)

13. Lincoln Lake can be roughly modeled as a semisphere with radius 1000 ft, and the flat circular part consisting of the surface of the lake.
    a. Calculate the volume of Lincoln Lake in ft^3. Use scientific notation. (1)
    b. Assume Lincoln Lake experiences an inflow from 15 inches of rain per year. Calculate its inflow from rain in ft^3/yr. (2)
    c. Assume Lincoln Lake's only outflow is through Lincoln Creek. You may model Lincoln Creek as described in the previous question. Calculate the outflow rate in ft^3/yr (2)
    d. Assume Lincoln Lake's total inflow is 4.0×10^{10} ft^3/yr. Will the lake level rise or fall? Explain why. (1)
    e. For this question only, assume Lincoln Lake is in volumetric equilibrium with the given inflow rate of 4.0×10^{10} ft^3/yr. What is its residence time in days? (2)

Refer to Figure B in the images sheet to answer the following questions.

14. Locate a sinkhole on the map. What are its approximate coordinates? (1)

15. While unmarked in this map, there are small streams that flow into the sinkhole. What are these called? (1)

16. What best describes the type of stream Stockton Valley Creek is? (1)
    a. Braided
    b. Meandering
    c. Straight
    d. Not enough information to determine
17. Many of the tributary river valleys contain extremely large, angular debris that is characteristic of the higher elevations in the area. This can be best explained due to formations of: (1)
    a. Sinkholes
    b. Springs
    c. Permafrost
    d. Solutional caves

18. Would carbonates be more strongly eroded by surface water or ground water? Briefly explain why. (2)

19. Calcium carbonate exhibits orthorhombic packing, which displays a primary porosity of 0.39.
    a. What does primary porosity mean? What does the value of 0.39 indicate? (2)

    b. Would calcium carbonate's secondary porosity be greater or lower than primary porosity? Briefly explain why. (2)

    c. Assume that calcium carbonate is instead packed cubically as shown in Figure C. Assume that the size of the calcium and carbonate ions are the same, spherical, and maximized. Calculate the primary porosity of this arrangement. Show all work! (3)

20. While unshown on the topographic map, the limestone is full of joints and fractures. As it erodes and widens, it becomes known as a... (1)
    a. Clint
    b. Grike
    c. Karren
    d. Fracture
21. What percentage of freshwater does soil moisture make up? (1)
    a. 0.05%
    b. 0.26%
    c. 0.03%
    d. 0.006%

22. True or False (and if false, explain why): The pore water pressure is greater than the atmospheric pressure in the vadose zone due to capillary forces. (2)

23. Pick one. Media with smaller pore sizes have <u>thicker/thinner</u> capillary fringes. Explain why smaller pores lead to <u>thicker/thinner</u> capillary fringes. (2)

24. Water in the unsaturated zone that flows laterally into a surface body of water, like a river, is called _____. (1)
    a. Overland Flow
    b. Baseflow
    c. Interflow
    d. Outflow
    e. None of the above
25. Interflow can be highly variable depending on the geology of a drainage basin. Which of these vadose zones geology would induce the most significant interflow. (1)
    a. Thin, permeable sandy layer over a fractured bedrock
    b. Thick, permeable sandy layer
    c. Thin layer of loam soil
    d. Thick layer of gravel over thin layer of loam soil

Refer to Figure D in the images sheet to answer the following questions.

26. Which letter refers to a feature that is common at the base of slopes in hilly terrain? What is its name? (2)

27. Which letter represents the point at which the water in a confined aquifer would reach if not trapped between confining layers? What is the name of this point? (2)

28. What features could possibly form when point A is just under the surface and/or intersecting the surface? Justify your answers using recharge rates. (2)

29. Why is location of point A significant in identifying effluent or influent streams? (1)

Refer to Figure E in the images sheet to answer the following questions.

30. What will occur if the precipitation rate is lower than the equilibrium infiltration capacity? (1)
    a. The precipitation will immediately pool at the surface.
    b. The precipitation will immediately infiltrate into the ground.
    c. The precipitation will infiltrate, but some precipitation will remain on the land surface.
    d. The precipitation will pool at the surface and gradually infiltrate into the ground.
    e. None of the above.
31. What will occur if the precipitation rate is greater than the equilibrium infiltration capacity but less than the initial infiltration capacity? (1)
    a. The precipitation will immediately pool at the surface.
    b. The precipitation will immediately infiltrate into the ground.
    c. The precipitation will infiltrate, but some precipitation will remain on the land surface.
    d. The precipitation will pool at the surface and gradually infiltrate into the ground.
    e. None of the above.

32. What will occur if the precipitation rate is greater than the initial infiltration capacity? (1)
    a. The precipitation will immediately pool at the surface.
    b. The precipitation will immediately infiltrate into the ground.
    c. The precipitation will infiltrate, but some precipitation will remain on the land surface.
    d. The precipitation will never infiltrate into the ground.
    e. None of the above.
33. Having a Manning roughness coefficient of 0.020 is most associated with what type of channel? (1)
    a. Smoothed concrete
    b. Winding natural streams with weeds
    c. Straight, unlined earth canal
    d. Mountain streams with rocky beds
34. In what situation would a gaining stream temporarily become a losing stream? (1)
    a. Drought
    b. Reduced ice melt
    c. Pumping groundwater
    d. Floods
35. Select the answer choice that lists the following processes in order. (1)

    1. Raindrops must reach sufficient size to ensure they will not evaporate before reaching the ground.
    2. Droplets coalesce to form raindrops.
    3. A humid air mass is cooled to dew-point temperature.
    4. Condensation of the nuclei is present.

    a. 4, 3, 2, 1
    b. 3, 4, 2, 1
    c. 3, 4, 1, 2
    d. 1, 2, 3, 4

Refer to Figure F in the images sheet to answer the following questions.

36. The area labeled by 1 refers to a zone where thawing occurs at some point in the year. What is the name of this zone? (1)

37. List three conditions for permafrost to be thicker. (3)

38. In the zone labeled by 2, the permafrost tends to be _____. (select all that apply) (1)
    a. Thicker
    b. Thinner
    c. Entirely Absent
39. If frozen ground can act as a low-conductivity aquitard, then the unfrozen ground beneath it can serve as a _____ _____, and the unfrozen active layer above it can be considered as a _____ _____. (1)
    a. Confined Aquitard; Confined Aquifer
    b. Unconfined Aquitard; Confined Aquifer
    c. Unconfined Aquifer; Confined Aquifer
    d. Perched Aquifer; Unconfined Aquifer
    e. Confined Aquifer; Perched Aquifer

40. Algal blooms in the Gulf of Mexico are commonly associated with high concentrations of phosphates and nitrates. What is the most likely cause? (1)
    a. Crude oil pollution
    b. Household chemicals pollution (bleach and detergent)
    c. Fertilizer runoff
    d. Algal blooms cannot be caused by phosphates and nitrates
41. How would increased water stratification in a lake affect its pollution? (1)
    a. It would be polluted longer due to less dilution
    b. It would be polluted for shorter due to more dilution
    c. Stratification cleans pollution by cycling water based on density
    d. None of the above are true

42. Which of the following generally have the longest pollutant residence time in a significant concentration? (1)
    a. Deep ocean
    b. Groundwater
    c. Lakes
    d. Streams
43. Which of the following best describes a critical zone? (1)
    a. Anything between the lowest limit of groundwater to the highest vegetation
    b. Regions at a critical tipping point of hydrological change
    c. Areas with ice features threatened by global warming
    d. Location identified to be important for human water usage
44. Which of the following best describes a saprolite? (1)
    a. Consolidated soil mass with vegetation
    b. Weathered bedrock with its original structure intact
    c. Unconsolidated soil mass with larger rocks
    d. Lightly fractured but unexposed bedrock

45. Total weathering rate W for a regolith of thickness C is modeled differently when thickness is above/below a certain critical thickness C_0, dependent on chemical weathering rate W_C and some constant k as follows:

$$\text{for } C < C_c, \quad W = W_c \left(1.0 + k\frac{C}{C_c} - \frac{C^2}{C_c^2} \right)$$

$$\text{for } C > C_c, \quad W = W_c \; k \; e^{-k(C - C_c)}$$

    a. How does the trend of weathering rate change when the critical point is reached? (1)

    b. When the system is purely mechanical weathering, disaggregation relieves stress on solid rock, leading to increased stability. In this case, would the critical thickness be high or low? (2)

    c. Are the changes in weathering rate at the critical thickness due mostly to chemical weathering or mechanical weathering? Explain how you know from the previous information. (2)

d. In most cases, weathering leads to greater porosity. How does this change the rate of chemical weathering? Explain why. (2)

46. Which of the following statements below does NOT describe the transition between laminar flow and turbulent flow? (1)
    a. The inertial forces of water movement become more influential than the viscous forces.
    b. The water particles begin to rush past each other in a rapid and organized fashion.
    c. The water particles stop moving along a parallel streamline.
    d. The moving fluid gains kinetic energy.
47. Darcy's Law is valid when the resistive forces of viscosity predominate, so oftentimes, Darcy's Law only applies to slowly moving water. Knowing this information, which of the following does NOT usually violate Darcy's Law? (1)
    a. Solution openings
    b. Basalt flows
    c. Pumping wells
    d. Groundwater
    e. All of them usually violate Darcy's Law
48. Refer to Figure G in the images sheet to answer the following questions.
    a. Which of the letters represents a lake which is being fed only by groundwater and drained by stream flow? (1)

    b. Which of the letters represents a reservoir created behind a dam? The water level created by this reservoir is _____ than the adjacent ground-water level. (1)

    c. The lake represented in stream F is a terminal lake. What is the primary way that water is discharged? (1)

    d. In the middle of Whoville is the lake represented by the letter A. Concerned about the bleak scenery, Carrie plants willow trees all around the border of the lake. How would the water table and lake inflow at the bottom be affected by Carrie's actions? (2)

    e. The town of Whoville currently has an outbreak of viral hepatitis, which is a water borne disease. Detective Annie pinpoints the source to a contaminated well. How was the well contaminated, and what is it contaminated by? (2)

    f. What is the role of the stagnation zone beneath a lake? (1)

49. True/False (If False, explain why): Hydrologists often refer to puddles as depression storage. (1)

50. In areas where the soil has a high infiltration rate, Horton overland flow occurs _____. (select all that apply) (2)
    a. During intense storms
    b. When the soil is saturated
    c. When the soil is frozen
    d. When the depression storage is filled

** End of the exam. Congratulations! **